We refer to the system of interest as the *active* system, or part A. The heat bath, or the bulk, is labelled B. Let us denote the total energy of the combined system AB by E. If the active system has energy $\mathrm{E_A}$ then the energy of the heat bath is $\mathrm{E_B} = \mathrm{E} - \mathrm{E_A}$. We suppose that the number of states accessible to a given system A or B depends on its energy, writing this as $Z_i(\mathrm{E}_i)$ where $i = \mathrm{A}$ or B. (For an idealized classical system, we may take Z_α to be the volume of the relevant energy surface.) As the size (number of particles) of the system is increased, the energy, being an *extensive* quantity, is expected to increase in proportion. We denote by $Z(\mathrm{E_A}, \mathrm{E_B})$ the total number of states the system can occupy with systems A and B having energies $\mathrm{E_A}$ and $\mathrm{E_B}$, respectively. We make use of our assumption that the two systems do not exchange particles so that the total number of states the combined system can be found in depends solely on the energies of the two independent systems, i.e.,

$$Z(\mathrm{E_A}, \mathrm{E_B}) = Z_\mathrm{A}(\mathrm{E_A}) \times Z_\mathrm{B}(\mathrm{E_B}).$$

Taking logarithms and replacing the value for $\mathrm{E_B}$ by $\mathrm{E} - \mathrm{E_A}$, we have

$$\ln Z(\mathrm{E_A}, \mathrm{E} - \mathrm{E_A}) = \ln Z_\mathrm{A}(\mathrm{E_A}) + \ln Z_\mathrm{B}(\mathrm{E} - \mathrm{E_A}).$$

The crucial assumption is that each of the states in $Z(\mathrm{E})$ is equally probable, and hence we can find the most likely energy distribution between the systems by maximizing the value of $Z(\mathrm{E_A}, \mathrm{E_B})$, or equivalently maximizing $\ln Z(\mathrm{E_A}, \mathrm{E} - \mathrm{E_A})$. This value of $\mathrm{E_A}$ occurs where

$$\frac{\mathrm{d}}{\mathrm{dE_A}} \ln Z(\mathrm{E_A}, \mathrm{E} - \mathrm{E_A}) = 0,$$

or equivalently where

$$\frac{\mathrm{d}}{\mathrm{dE_A}} S_\mathrm{A} = \frac{\mathrm{d}}{\mathrm{dE_B}} S_\mathrm{B},$$

where we have used Boltzmann's relation to express this in terms of the entropy.

We have assumed that our systems are in thermal equilibrium, but if we had not, and instead invested the heat bath with the total system energy, we would expect that the energy would transfer back into system *A* until the point where the stationarity relation is satisfied. This assumption is the same as saying that the system obeys the second law of thermodynamics; furthermore this equilibrium occurs when the entropy of the system is at its maximum.

The temperature of a system T, is defined as

$$\frac{1}{\mathrm{T}} := \frac{\partial S}{\partial \mathrm{E}},$$

giving, when the system is in thermal equilibrium,

$$\frac{\mathrm{d}}{\mathrm{dE_A}} \ln Z_\mathrm{A}(\mathrm{E_A}) = \frac{\mathrm{d}}{\mathrm{dE_B}} \ln Z_\mathrm{B}(\mathrm{E_B}) = \frac{1}{\mathrm{k_B T}} =: \beta. \tag{6.2}$$

We shall refer to the quantity β as the system's inverse or reciprocal temperature.

The volume of states of the active system with energy α is $Z_\mathrm{A}(\alpha)$. On the other hand the probability for the active system to have energy $\leq \alpha$ is the same as the probability of the bath having energy $\geq \mathrm{E} - \alpha$, this must be

$$P(\alpha) = \Pr\{\mathrm{E_B} \geq \mathrm{E} - \alpha\} \propto \int_0^\alpha Z_\mathrm{B}(\mathrm{E} - s)\mathrm{d}s.$$

Since $Z_\mathrm{B}(\mathrm{E} - s) = \exp(S_\mathrm{B}(\mathrm{E} - s)/\mathrm{k_B})$, and $s \ll \mathrm{E}$, we (naively[3]) apply the mean value theorem to S_B to obtain

$$S_\mathrm{B}(\mathrm{E} - s) = S_\mathrm{B}(\mathrm{E}) - S'_\mathrm{B}(\xi)s,$$

and from $S'_\mathrm{B}(\xi) = 1/\mathrm{T}$, we obtain

$$Z_\mathrm{B}(\mathrm{E} - s) \propto \exp(-\beta s),$$

where $\beta = (\mathrm{k_B T})^{-1}$. Thus we have

$$P(\alpha) \propto \int_0^\alpha \exp(-\beta s)\mathrm{d}s.$$

This is the probability of finding any given active system of our ensemble having energy $\leq \alpha$. The volume (or number, in the case of discrete systems) of states available at energy α is $Z_\mathrm{A}(\alpha)$. Thus the energy density of the canonical ensemble at energy level α is proportional to $Z_\mathrm{A}(\alpha)\exp(-\beta\alpha)$.

6.1.3 The Gibbs-Boltzmann Probability Density

Observe that the previous discussion does not quite tell us the form of the phase space density associated to the canonical ensemble. An elegant way to derive this

[3]This is one of several imprecisions in this discussion whose correction would entail significant additional labor; since our ultimate aim is to use the framework and not to provide its rigorous derivation, we prefer to retain the intuitive continuity of presentation, in keeping with popular treatments.